

Security Assessment: Metasploit & Nmap Reconnaissance

1. Executive Summary

In this report, I walk through a simulated reconnaissance exercise using the **Metasploit Framework** and its integration with **Nmap** to assess a local target system's network security posture. The objective was to detect open ports, identify running services, and determine potential vulnerabilities.

The result? **No open ports, no running services, and no detected vulnerabilities** — indicating a well-secured system.

2. Background and Objectives

This project simulates the **initial phase of a penetration test: reconnaissance**. The goal was to use Metasploit's **db-nmap** command (which integrates Nmap) to perform a detailed scan of a target machine on a local network.

Why Metasploit + Nmap?

Metasploit provides a powerful environment to manage and store scan results, while Nmap is one of the most reliable tools for discovering network services.

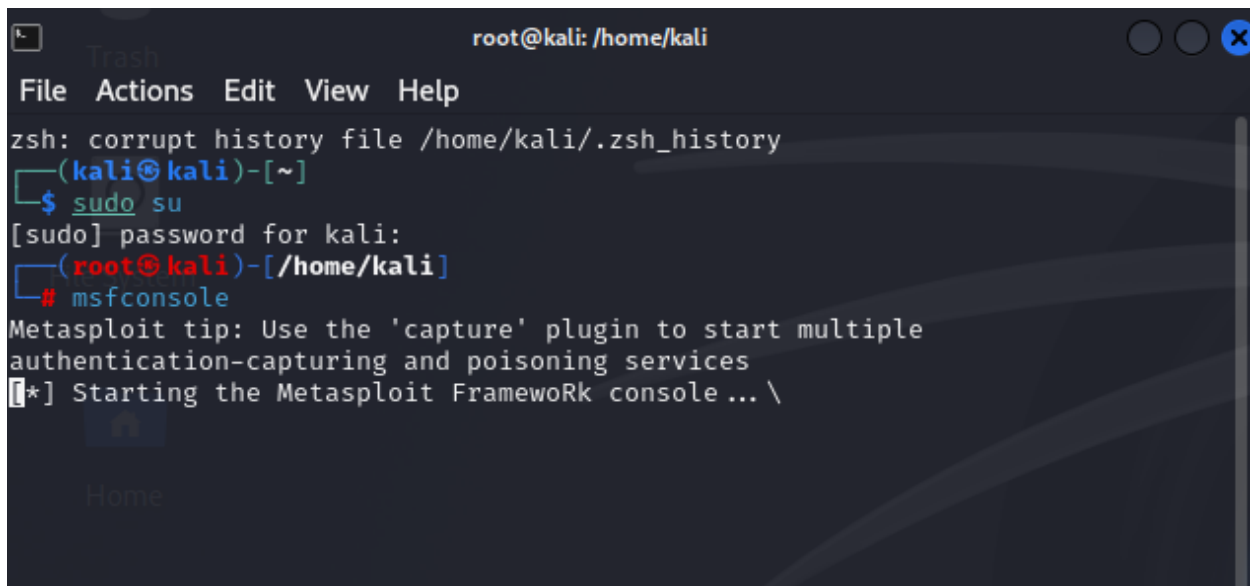
3. Methodology

3.1 Tool Configuration

Operating system used: Kali Linux

The setup involved a few key steps:

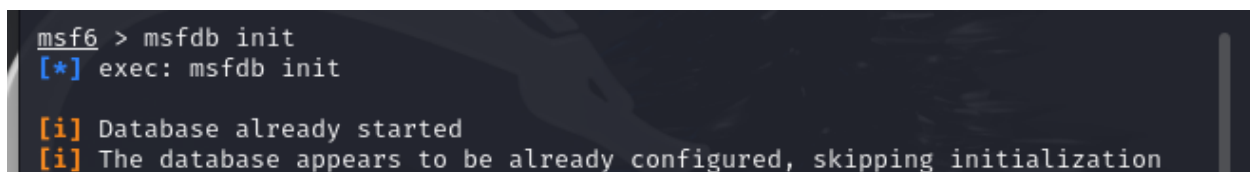
- Gained root access using: **Sudo su**
- Launched Metasploit with: **msfconsole**



```
root@kali: /home/kali
File Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(root@kali)-[/home/kali]
# msfconsole
Metasploit tip: Use the 'capture' plugin to start multiple
authentication-capturing and poisoning services
[*] Starting the Metasploit Framework console ... \
```

*Initializing the Metasploit environment: **msfconsole** opens the framework console*

- Initialized the database: **msfdb init**



```
msf6 > msfdb init
[*] exec: msfdb init

[i] Database already started
[i] The database appears to be already configured, skipping initialization
```

***msfdb init** sets up the backend database for session tracking and Nmap integration*

- Identified the target IP using: **ifconfig (on kali Linux)**
- Ran the scan: **db_nmap -sV -sC -vv [Target_IP]**

where **-sC** is for Default NSE scripts

- **-sV** is to Detect service versions
- **-vv** is to Enable verbose output

3.2 Execution Process

Using the **db-nmap** command within the Metasploit console, the scan focused on identifying exposed services, ports, and service versions. The Nmap engine ran 156 built-in scripts to evaluate the target.

3.3 Monitoring and Analysis

The output showed:

- **All 1000 TCP ports closed**
- **No services identified**
- **156 scripts executed, no vulnerabilities found**

This implies the system had **no accessible attack surface** during the scan.

4 Findings and Analysis

4.1 Indicators of Compromise (IOCs)

None detected. The system appeared clean and unexposed.

4.2 Behavioral Observations

- **Closed ports across the board**
- **No active services**
- **Script-based scans returned no actionable results**

The lack of visible services suggests **strong security configurations**, possibly due to firewall rules or host-based security settings.

```
[*] Nmap: Not shown: 1000 closed tcp ports (reset)
[*] Nmap: NSE: Script Post-scanning.
[*] Nmap: NSE: Starting runlevel 1 (of 3) scan.
[*] Nmap: Initiating NSE at 07:40
[*] Nmap: Completed NSE at 07:40, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 2 (of 3) scan.
[*] Nmap: Initiating NSE at 07:40
[*] Nmap: Completed NSE at 07:40, 0.00s elapsed
[*] Nmap: NSE: Starting runlevel 3 (of 3) scan.
[*] Nmap: Initiating NSE at 07:40
[*] Nmap: Completed NSE at 07:40, 0.00s elapsed
[*] Nmap: Read data files from: /usr/share/nmap
[*] Nmap: Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
[*] Nmap: Nmap done: 1 IP address (1 host up) scanned in 0.84 seconds
[*] Nmap: Raw packets sent: 1000 (44.000KB) | Rcvd: 2000 (84.000KB)
msf6 > 
```

Nmap scan result indicating that all 1000 TCP ports on the target system are closed, with no exposed services detected.

4.3 Risk and Impact Assessment

The absence of open ports and services reduces the potential risk from external attackers. While no issues were found, it's important to note:

- **No vulnerabilities found" ≠ "No vulnerabilities exist"**
- There could be **hidden services** or **non-TCP based threats** not exposed via this method

5 Recommendations

5.1 Immediate Actions

- **Audit firewall rules** to confirm intentional port closure
- **Validate that defensive measures** like service cloaking aren't unintentionally blocking essential operations

5.2 Long-Term Mitigation

- Schedule **regular vulnerability scans**
- Use **alternative tools** or **OSINT techniques** to deepen assessments
- Incorporate **continuous monitoring** to adapt to new threats

6. Conclusion

The scan conducted with Metasploit's db_nmap tool revealed no exploitable entry points — a promising sign for the system's overall security.

Key Takeaway:

Sometimes, the absence of findings is itself a valuable finding — it shows that layered defenses are working.

Next Steps

- Continue regular scans
- Perform **deeper inspection with diverse tools**
- Stay proactive about patch management and configuration hardening